

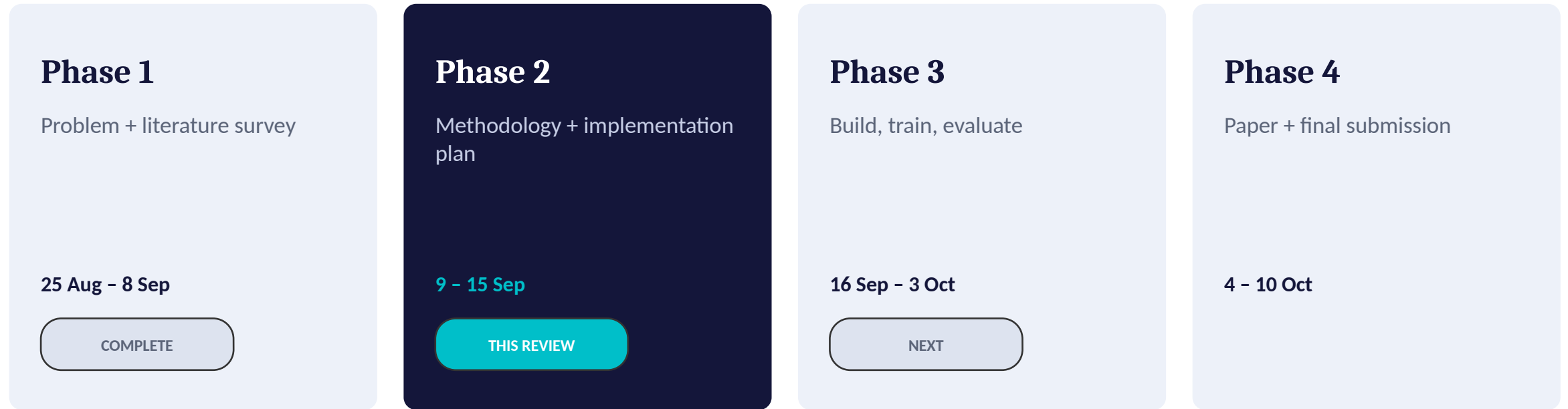
PHASE 2 • 1ST REVIEW

Deep Learning–Based Neural Decoder for EEG Motor Imagery

Methodology and Implementation Plan

15 September 2026

Where this review sits



The 0th Review established the problem and surveyed four architectures. This review commits to a specific method, a testable hypothesis, and a dated plan to produce results by 3 October.

THE PROBLEM, IN ONE NUMBER

82.52%

on people it was
trained on



58.64%

on a person it
has never met

-23.88

percentage points lost —
same architecture, same dataset,
only the test subject changed

A decoder that scores 82% on people it has met and 59% on a stranger is not a deployable assistive device. A patient cannot be asked to supply hours of labelled calibration data before their wheelchair responds.

Three constraints that pull against each other

1

High accuracy

on subjects never seen during training — not just on the people in the training set

2

Lightweight

small enough to run on low-power assistive hardware with low latency

3

No lengthy calibration

usable without hours of labelled data from each new user

The unresolved question

Recent architectures close the cross-subject gap by adding capacity — attention layers, transformer depth, larger parameter counts — which works against constraints 2 and 3. Nobody has established whether the gain comes from architectural capacity, or from making the input representation subject-invariant in the first place.

OUR HYPOTHESIS

H1

Subject-invariance at the input contributes more to cross-subject decoding than an equivalent increase in model capacity — and the two are complementary, not redundant.

H2

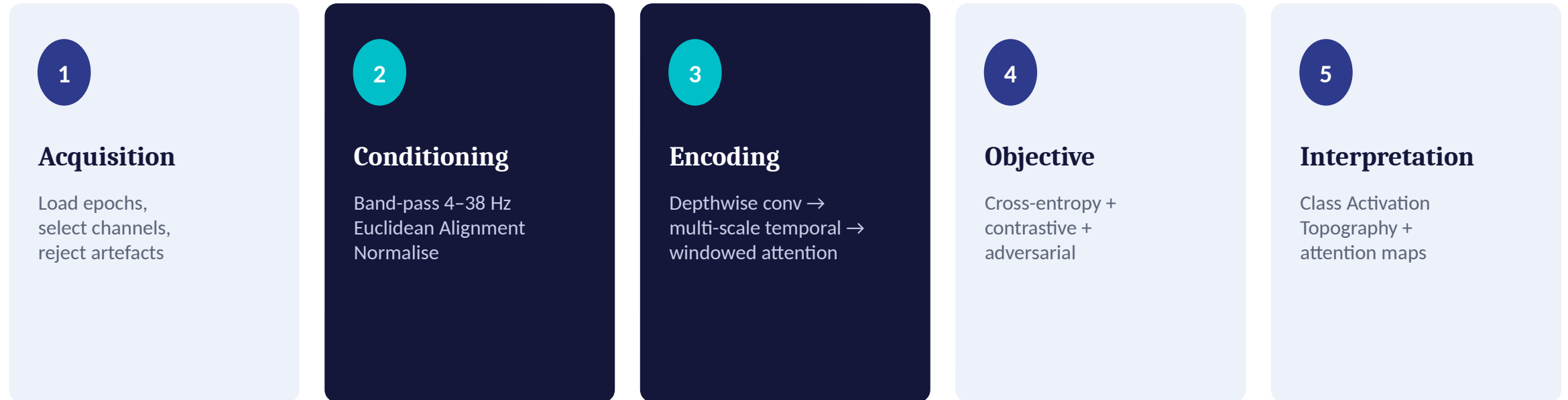
Windowed attention matches full self-attention under LOSO with fewer parameters — unconstrained capacity overfits subject-specific structure on ~2,600 trials.

H3

Input alignment reduces the number of labelled trials a new user must supply to reach a fixed accuracy.

Every hypothesis here is falsifiable by a specific experiment in Section 8. If H1 is wrong, that is a reportable finding — it would tell the field that architectural scaling is the productive direction.

Five-stage pipeline



Stages 2 and 3 carry the contribution. Stage 2 removes subject-specific covariance before the network ever sees the data; stage 3 is the lightweight hybrid encoder.

SAND-Net: every component traces to a paper

Subject-Agnostic Neural Decoder — nothing is included because it appeared in a paper; each component must earn its place in the ablation.

Component	Taken from	Why it is included
Euclidean Alignment	He & Wu (2020)	Removes subject covariance shift — unsupervised, needs no target labels
Depthwise-separable conv	EEGNet (2018)	CSP-like spatial filters at very low parameter cost
Multi-scale dilated conv	ATCNet (2023) / CTNet (2024)	Mu and beta rhythms evolve on different timescales
Windowed self-attention	EEG Conformer (2023)	Long-range dependency without $O(T^2)$ cost or overfitting
Contrastive + adversarial loss	Domain-generalisation lit.	Forces class structure to be shared across subjects
Class Activation Topography	EEG Conformer (2023)	Channel contribution per class, for physiological validation

Highlighted row: the one addition none of the four reviewed architectures has. Target budget under 50,000 parameters — between ShallowConvNet (47,364) and ATCNet (113,732).

The calibration-free mechanism

Euclidean Alignment

$$\mathbf{R}_s = 1/N \cdot \sum \mathbf{X}_i \mathbf{X}_i^T$$

$$\tilde{\mathbf{X}}_i = \mathbf{R}_s^{-1/2} \mathbf{X}_i$$

After the transform, every subject's mean covariance equals the identity matrix.

Unsupervised

Needs only unlabelled trials from the new user — never their labels

Nearly free

One eigendecomposition of a 22×22 matrix; negligible against a forward pass

Deployable

New user records a minute of unlabelled imagery; the matrix is computed from it

This is what makes "without lengthy calibration" operational rather than aspirational. Most recent work attacks subject variability inside the network. We attack it before the network sees the data — and then test, in the ablation, which of the two actually matters.

Three motor imagery datasets, tiered

PRIMARY

BCI Competition IV-2a

9 subjects · 22 channels
4 classes · 250 Hz
576 trials/subject

Common to all four reviewed papers — lets us sanity-check reproduced baselines

SECONDARY

BCI Competition IV-2b

9 subjects · 3 channels
2 classes · 250 Hz
720 trials/subject

Only C3, Cz, C4 — tests survival under drastic loss of spatial information

STRETCH

High Gamma Dataset

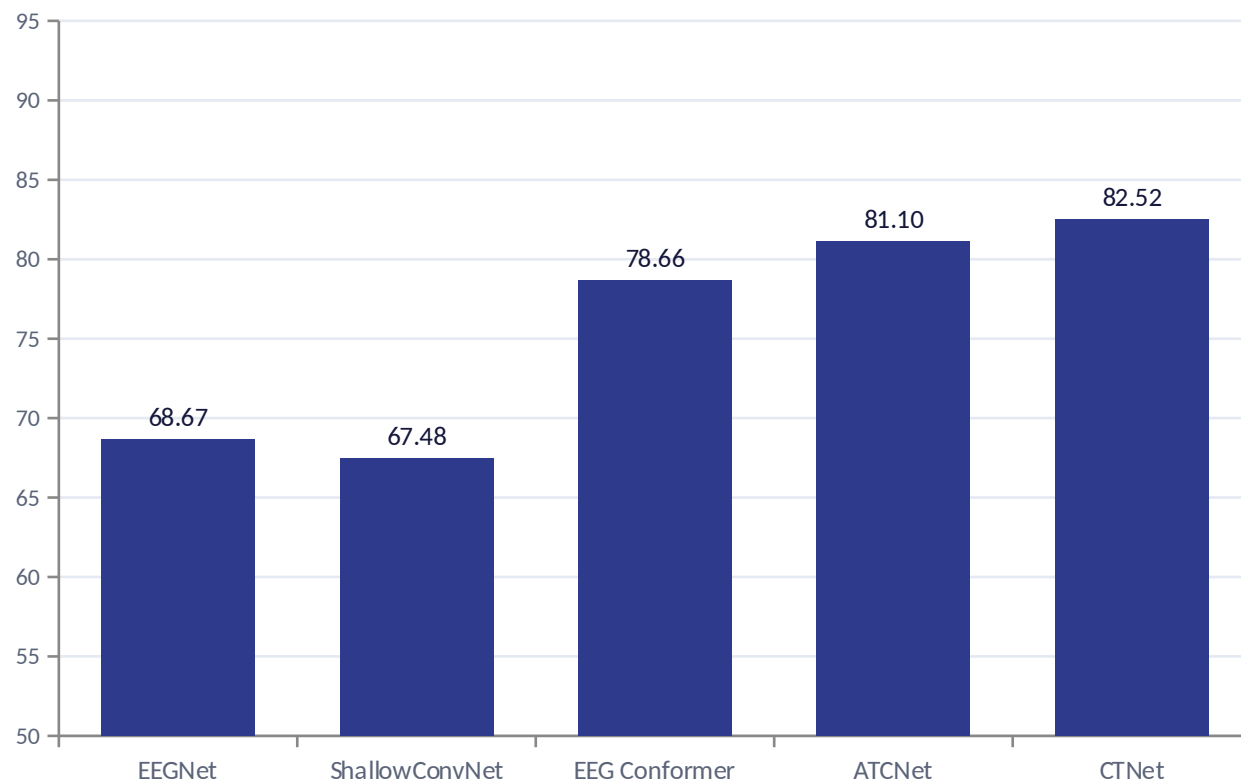
14 subjects · 128 channels
4 classes · 500 Hz
~880 trials/subject

Harder generalisation test; attempted only if the schedule allows

SEED is deliberately excluded

EEG Conformer also evaluates on SEED, but SEED is emotion recognition — participants watch film clips and the label is positive / neutral / negative. It contains no motor imagery and no movement labels. Training jointly across two unrelated label spaces would measure nothing about motor intent decoding.

Published numbers cannot be ranked against each other



Different protocols

Hold-out, 5-fold, session-based and subject-specific splits all appear in this chart.

Same model, two figures

EEGNet is quoted at 68.67% by the ATCNet authors and 71.50% by another — same model, same dataset.

All within-subject

These are not cross-subject numbers. The cross-subject picture is much lower.

Our Objective 2

Re-run all four under one identical LOSO protocol. That comparison does not exist in the literature.

Leave-one-subject-out, with a held-out validation subject

Fold	Training (7 subjects)	Validation (1)	Test (1)
1	S3 – S9	S2	S1
2	S1, S4 – S9	S3	S2
...	...	...	...
9	S2 – S7	S1	S9

Reported figure is the mean and standard deviation across all 9 folds.

Why a validation SUBJECT, not a validation split

Validating on trials from subjects that also appear in training selects hyperparameters that favour within-subject fit — exactly the behaviour we are trying to avoid.

Leakage control

Normalisation statistics, augmentation and hyperparameter choices come only from training and validation subjects. The test subject contributes unlabelled trials to alignment — the same information a real deployment would have.

Metrics: accuracy (mean $\pm$ std), Cohen's kappa, macro-F1, confusion matrix, parameters / FLOPs / latency, and a Wilcoxon signed-rank test paired across the 9 subjects at $\alpha = 0.05$.

The ablation is the experiment that tests H1

	EA	Multi-scale	Wind. attn	SupCon	DANN	What this row is
A1	—	—	—	—	—	EEGNet-equivalent baseline
A2	✓	—	—	—	—	Alignment alone
A3	—	✓	✓	—	—	Capacity alone
A4	✓	✓	—	—	—	Alignment + multi-scale
A5	✓	✓	✓	—	—	Full architecture, no DG losses
A6	✓	✓	✓	✓	—	Plus contrastive
A7	✓	✓	✓	—	✓	Plus adversarial
A8	✓	✓	✓	✓	✓	Full SAND-Net

A1 vs A2 vs A3

The core test of H1. A2 isolates alignment with no added capacity; A3 isolates capacity with no alignment.

A3 vs A5

Does alignment still help once the architecture is already strong?

A5 vs A6 / A7 / A8

Are the domain-generalisation losses complementary to input alignment, or redundant with it?

A4 vs A5

Does windowed attention earn its parameters?

Each configuration runs over all 9 LOSO folds — 72 training runs. If $A2 - A1 > A3 - A1$, H1 is supported.

Two experiments the reviewed papers do not run

1

Calibration efficiency

After LOSO training, fine-tune on n labelled trials from the held-out subject for $n = 0, 5, 10, 20, 40, 80$ and plot accuracy against n .

A model that becomes usable at $n = 10$ is far more deployable than one needing $n = 80$ — even if their accuracies at $n = 80$ are identical. This is the figure tied most directly to the clinical motivation.

2

Neurophysiological validation

Motor imagery produces event-related desynchronisation contralateral to the imagined limb: left-hand imagery suppresses mu rhythm near C4, right-hand near C3.

PASS — contralateral peaks at C3 / C4: the model learned real physiology.

FAIL — frontal or symmetric peaks: the model is reading ocular or muscular artefacts, and the accuracy figure is not measuring motor intent regardless of how high it is.

Interpretability here is a correctness check, not a presentation aid. It has a real failure mode and we will report the result either way.

Phase 3: 18 days to results

16–17 Sep	Environment and data	All three datasets downloading; EDA plots produced
18–19 Sep	Preprocessing + alignment	Cached tensors; alignment verified (covariance $\approx$ identity)
20–22 Sep	Baselines, within-subject	EEGNet and Conformer reproduce published accuracy within ± 3 pts
23–25 Sep	LOSO harness + all baselines	9-fold LOSO results for all five baselines on IV-2a
26–28 Sep	SAND-Net implementation	Trains end to end; parameter count under 50k confirmed
29–30 Sep	Ablation study	All 8 configurations $\times$ 9 folds complete
1–2 Oct	IV-2b, calibration, interpretability	Cross-dataset results; topographies; calibration figure
3 Oct	Consolidation $\rightarrow$ 2nd Review	Final tables, figures and significance tests

Front-loaded deliberately: the pipeline and baselines must work before the proposed model is written. A proposed model evaluated on a broken pipeline is worthless.

The plan is only credible if it fits the GPU budget

Compute estimate

Experiment set	Runs	Est. total
Baselines, LOSO, IV-2a	45	≈ 6 h
SAND-Net, LOSO, IV-2a	9	≈ 1.2 h
Ablation, 8 configs	72	≈ 9.6 h
IV-2b, all models	54	≈ 2.7 h
Calibration curve	54	≈ 1.8 h
Contingency	—	≈ 5 h
Total		≈ 26 GPU-hours

Kaggle provides roughly 30 GPU-hours per week, so this fits — provided preprocessing is cached and every run is checkpointed.

Top risks, with the response already decided

Runtime exceeds GPU hours

Cache tensors, mixed precision, drop the High Gamma tier first

Baselines miss published accuracy

Use Braindecode / official repos; report both figures and explain the gap

Session timeouts

Checkpoint every epoch; resumable trainer

SAND-Net does not beat baselines

The ablation still answers H1 — a negative result is a valid finding

WHAT WE WILL SHOW ON 3 OCTOBER

Success criteria, fixed in advance

$\geq 70\%$

LOSO accuracy on IV-2a

Independent LOSO studies place
EEGNet and Conformer at 68–70%

$< 50k$

parameters

Below ShallowConvNet (47,364), well
below ATCNet (113,732)

≤ 20

labelled trials to calibrate

Within 5 points of subject-specific
accuracy

$p < 0.05$

Wilcoxon signed-rank

Paired across the 9 subjects — the test
Conformer uses

On novelty

The 0th Review answered “Partially”. That investigation is now complete. The contribution is not a combination of existing techniques — it is the first evaluation of all four architectures under one identical LOSO protocol, a controlled test of whether input alignment or architectural capacity drives cross-subject decoding, and a calibration-efficiency characterisation stating how much labelled data a new user actually needs. The last two are questions the reviewed literature does not answer, and both yield a useful result whichever way they resolve.